Rebecca M. M. Hicke

PhD Candidate, Computer Science, Cornell University

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Education

Cornell University, Ph.D. Candidate, Computer Science — *(expected)* May 2027
Minor: Information Science — *Ithaca, NY*
Committee: David Mimno (chair), Lillian Lee, Jon Kleinberg, Ross Kristensen-McLachlan
Thesis areas: Computational literary studies, natural language processing (NLP)

Cornell University, M.S., Computer Science — December 2024
Minor: Information Science — *Ithaca, NY*

Carleton College, B.A. *cum laude*, Computer Science and English — June 2022
Minor: Digital Arts and Humanities — *Northfield, MN*
Received distinction in both majors

Research & Industry Experience

Graduate Research Assistant, Cornell University — August 2022 – present
Advisor: David Mimno — *Ithaca, NY*
Areas: Computational literary studies, NLP

AI Research Scientist, AIIL Group, Microsoft Research — November 2025 – January 2026
Advisor: Kiran Tomlinson — *Remote*
Areas: Human-AI interaction, NLP

Research Intern, AIIL Group, Microsoft Research — May 2025 – August 2025
Advisor: Kiran Tomlinson — *Redmond, WA*
Areas: Human-AI interaction, NLP

Graduate Research Assistant, Aarhus University — May 2024 – August 2024
Advisor: Ross Deans Kristensen-McLachlan — *Aarhus, Denmark*
Areas: Computational literary studies, NLP

PiTech Research Fellow, Cornell University — June 2023 – August 2023
Host Organization: Good Call NYC — *Remote*
Areas: Data visualization, data analytics

Research Scientist, Authors AI — May 2023 – August 2023
Areas: NLP — *Remote*

Undergraduate Research Assistant, Carleton College — June 2020 – May 2022
Advisor: Eric Alexander — *Northfield, MN*
Areas: Data visualization, Human-computer interaction (HCI), digital humanities

Undergraduate Research Assistant, UC Berkeley — June 2021 – March 2022
Advisors: Sarah Chasins and Aditya Parameswaran — *Remote*
Areas: HCI, NLP

Teaching Experience

Instructor of Record, Cornell University — June 2026 – August 2026
INFO 1100: Introduction to Programming
Designed course and materials (available on my website)

Research Group Instructor and Mentor, Cornell University — Spring 2024 – Spring 2025
Content Analysis and Large Language Model Assisted Narrative Research
Co-designed and directed research course, mentored 8 students

Guest Lecturer
Text Mining History and Literature, Cornell University — August 2026
taught by David Mimno
NLP for Cultural Analytics, University of Colorado Boulder — September 2025
taught by Maria Antoniak
Natural Language Processing, Carleton College — October 2024
taught by Eric Alexander
How LLMs Work: Their Potential and Limitations, Cornell University — October 2024
taught by David Mimno

Graduate Teaching Assistant, Cornell University — August 2023 – present
INFO 3350: Text Mining History and Literature (1 term)
CS/INFO 4300: Language and Information (1 term)
Received Outstanding Graduate Teaching Assistant Award
CS 2110: Object-Oriented Programming and Data Structures (1 term)
CS/INFO 3300: Data-Driven Web Applications (1 term)

Undergraduate Teaching Assistant, Carleton College — March 2020 – June 2022
CS 111: Introduction to Computer Science (1 term)
CS 201: Data Structures (4 terms)
CS 202: Math of Computer Science (2 terms)

Publications

** Represents undergraduate mentees at the time of publication*

Peer-Reviewed Conference and Journal Publications

Hicke, Rebecca M. M. and Kiran Tomlinson. **Adopt ≠ Adapt: Longitudinal Analyses of LLM Conversations in the Wild**. *Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)*.

Hicke, Rebecca M. M., Brian W. Haggard, Mia Ferrante,* Rayhan Khanna,* and David Mimno. **Are You There God? Lightweight Narrative Annotation of Christian Fiction with LMs**. *Conference on Computational Humanities Research (CHR 2025)*.

Hicke, Rebecca M. M., Yuri Bizzoni, Pascale Feldkamp, and Ross Deans Kristensen-McLachlan. **Says Who? Effective Zero-Shot Annotation of Focalization**. *Conference on Computational Humanities Research (CHR 2025)*.

Hicke, Rebecca M. M., Sil Hamilton, and David Mimno. **The Zero Body Problem: Probing LLM Use of Sensory Language**. *Conference on Language Modeling (COLM 2025)*.

Hicke, Rebecca M. M. and David Mimno. **Looking for the Inner Music: Probing LLMs' Understanding of Literary Style**. *Expanding the Toolkit: Large Language Models in Humanities Research in Computational Humanities Research (2025)*.

Hicke, Rebecca M. M. and Ross Deans Kristensen-McLachlan. **SCIENCE IS EXPLORATION: Computational Frontiers for Conceptual Metaphor Theory**. *Conference on Computational Humanities Research (CHR 2024)*.
Finalist – Best Short Paper Award

Kristensen-McLachlan, Ross Deans, Rebecca M. M. Hicke, Márton Kardos, and Mette Thunø. **Context is Key(NMF): Modelling Topical Information Dynamics in Chinese Diaspora Media**. *Conference on Computational Humanities Research (CHR 2024)*.

Hicke, Rebecca M. M. and David Mimno. **T5 meets Tybalt: Author Attribution in Early Modern English Drama using Large Language Models**. *Conference on Computational Humanities Research (CHR 2023)*.

Book Chapters

Hicke, Rebecca M. M. and David Mimno. **'Let every word weigh heavy of her worth': Examining How Women Enact Power in Shakespeare's Comedies through Interactive Speech Pattern Visualizations**. In *Computational Drama Analysis: Reflecting Methods and Interpretations (*2024*)*. De Gruyter.

Workshop Publications

Hamilton, Sil, Rebecca M. M. Hicke, Mia Ferrante,* David Mimno, and Matthew Wilkens. **Too Long, Didn't Model: Decomposing LLM Long-Context Understanding with Novels**. *Joint SIGHUM Workshop on Computational Linguistics for Cultural Heritage, Social Sciences, Humanities and Literature (LaTeCH-CLfL 2026)*.

Hamilton, Sil, Rebecca M. M. Hicke, Matthew Wilkens, and David Mimno. **A City of Millions: Mapping Literary Social Networks at Scale**. *International Conference on Natural Language Processing for Digital Humanities (NLP4DH 2025)*.

Hicke, Rebecca M. M. and David Mimno. **[Lions: 1] and [Tigers: 2] and [Bears: 3], Oh My! Literary Coreference Annotation with LLMs**. *Joint SIGHUM Workshop on Computational Linguistics for Cultural Heritage, Social Sciences, Humanities and Literature (LaTeCH-CLfL 2024)*.

Hicke, Rebecca M. M., Maanya Goenka, and Eric Alexander. **Word Clouds in the Wild**. *IEEE Workshop on Visualization for the Digital Humanities (Vis4DH 2022)*.

Under Review

Hicke, Rebecca M. M., Sil Hamilton, David Mimno, and Ross Deans Kristensen-McLachlan. **The Plot Thins: Uniformity and Linearity in Literary Summaries**. *Under review at CHR 2027*.

Bax, Axel, Sanghoon Oh, Rebecca M. M. Hicke, Matthew Wilkens, and David Mimno. **Judging a Cover by its Book**. *Under review at CHR 2027*.

SERVICE

Workshop Organizer: NLP for the Digital Humanities [NLP4DH] (2025)

Session Chair: CHR (2023)

Reviewer: LaTeCH-CLfL (2024, 2025, 2026), DH (2024, 2025, 2026), CHR (2023, 2027), CHR Journal (2025), Open Research Europe (2023), Journal of Computational Literary Studies (2023)

Mentoring and Outreach:

ACSU Research Night (2023, 2024, 2025)
Recruited graduate presenters for outreach getting students interested in research

Student-Applicant Support Program (2022, 2023, 2024, 2025)
Provided feedback on application materials for applicants to Cornell CS PhDs

New Visions in Engineering (2022-2023, Coordinator: 2023-2024)
Mentored high school students interested in futures in engineering + CS research

Bowers Undergraduate Research Experience (BURE) (2023)
Mentored and organized social events for undergraduate research assistants

Program Assistant, Carleton College (June 2022 – August 2022)
Mentored and supervised students in History of Computing Study-Abroad Program

Residential and Program Assistant at UT Austin (2018 – 2020)
Academy for Women – formerly First Bytes (2 sessions)
Academy for All – formerly Code Longhorn (1 session)
Academy for Robotics (2 sessions)
Academy for Cybersecurity (1 session)

Awards & Fellowships

Future Leaders in Liberal Arts AI Education Travel Grant, *Harvey Mudd College*	May 2026
Foundational AI PhD Fellowship, *Cornell University*	January 2026 – May 2026
Hopcroft Fellowship, *Cornell University*	2022 – 2023
Laurence McKinley Gould Prize in Natural Science, *Carleton College*	May 2022
David John Field Prize, *Carleton College*	May 2022
The David Pollatsek '96 Prize in Computer Science, *Carleton College*	May 2022
Distinction – Computer Science Senior Thesis, *Carleton College*	May 2022
Distinction – English Senior Thesis, *Carleton College*	April 2022
Thorsen Scholarship, *Carleton College*	2020 – 2022
Larson International Fellowship, *Carleton College*	2021

Certifications & Supplemental Training

(Re)Imagining Liberal Arts & STEM Education in the Age of GenAI, Harvey Mudd College — May 2026
Conference on adapting education in response to the proliferation of AI

Early Career Graduate Teaching Cohort, Cornell University — Spring 2026
Workshop series on developing teaching skills

Essentials of Teaching Institute, Cornell University — Fall 2025
Course on developing teaching skills

PRESENTATIONS

Invited Talks

"Annotation as AI Pedagogy," — May 2026
(Re)Imagining Liberal Arts and STEM Education in the Age of GenAI

"Says Who? Effective Zero-Shot Annotation of Focalization," — April 2026
Quid deinde? New Ideas and Approaches to AI & DH, University of Montreal

"How to Read Books with AI: AI and Narrative Analysis," *University of Richmond* — March 2026

Conference Talks

"Says Who? Effective Zero-Shot Annotation of Focalization," *CHR 2025* — December 2025

"Are You There God? Lightweight Narrative Annotation of Christian Fiction with LMs," *CHR 2025* — December 2025

"SCIENCE IS EXPLORATION: Computational Frontiers for Conceptual Metaphor Theory," *CHR 2024* — December 2024

"T5 meets Tybalt: Author Attribution in Early Modern English Drama Using Large Language Models," *CHR 2023* — December 2023

Workshop Talks

"Word Clouds in the Wild," Vis4DH @ IEEE Vis — October 2022

"Examining How Women Enact Power in Shakespeare's Comedies through Interactive Speech Pattern Visualizations," *Workshop on Computational Drama Analysis* — September 2022

Poster Presentations

"The Zero Body Problem: Probing LLM Use of Sensory Language," *COLM 2025* — October 2025

"[Lions: 1] and [Tigers: 2] and [Bears: 3], Oh My! Literary Coreference with LLMs," *LaTeCH-CLfL 2024* — March 2024

"T5 meets Tybalt: Author Attribution in Early Modern English Drama Using Large Language Models," *TADA 2023* — November 2023

"Large Language Models and NER: Better Results with Less Work," *DH 2023* — July 2023